

Faster Algebraic Shifting

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Simplicial Complexes

Definition

An abstract simplicial complex K , is a collection of non empty sets of $[n] := \{1, 2, \dots, n\}$, such that for any $\tau, \sigma \subset [n]$, if $\tau \subset \sigma \in K$ then $\tau \in K$.

We call $\sigma \in K$ a k -simplex or a face of dimension k if $|\sigma| = k + 1$.

The f -vector counts the number of k -faces for each k .

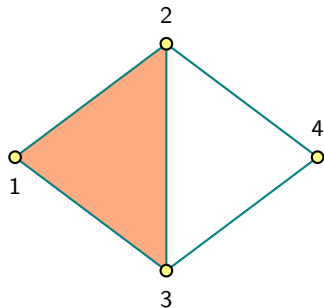
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The f -vector is $(4, 5, 1)$

Simplicial Homology

Definition

Given a field $\mathbb{F}$, and a simplicial complex K we define the k th simplicial homology to be the vector space

$$H_k^{\mathbb{F}}(K) := k\text{-cycles} / k+1\text{-boundaries}.$$

The k th Betti number of K is the dimension of $H_k^{\mathbb{F}}(K)$.

Simplicial Homology

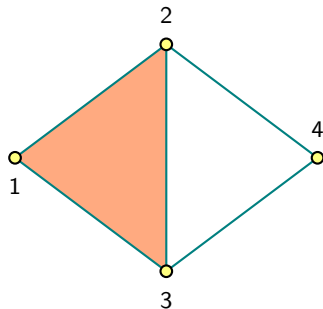
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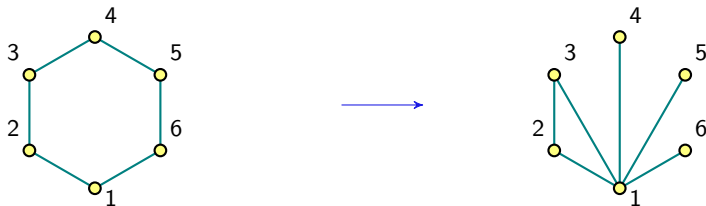
The Betti numbers are $(1, 1, 0)$



Algebraic Shifting

Definition

A k -uniform hypergraph is collection $C \in \binom{[n]}{k}$. C is shifted if for any $\sigma \in C$, $\sigma|_i^j \in C$ for $i \in \sigma$ and $j < i$.

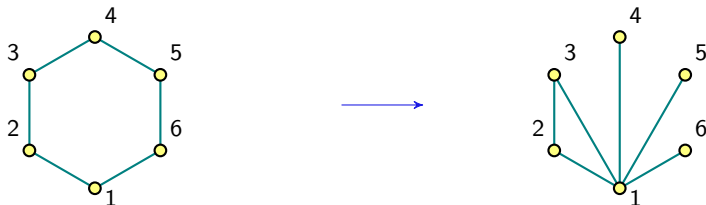


The k -simplices of a simplicial complex form a k -uniform hypergraph.

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Theorem (Kalai '84, Björner-Kalai '88)

Given K a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(K)$ with the same f -vector and Betti numbers as K .

Compound Matrices

Definition

Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \text{GL}(n, \mathbb{E})$.

The k th compound matrix $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$ -matrix of k -minors of g .

$g^{\wedge K_k}$ is the $|K_k| \times \binom{n}{k}$ -submatrix of $g^{\wedge k}$ with rows indexed by K_k .

Let $\sigma_i \in K_k$, $|K_k| = m$ and $\tau_i \in \binom{[n]}{k}$ such that

$$\sigma_1 <_{\text{lex}} \sigma_2 <_{\text{lex}} \cdots <_{\text{lex}} \sigma_m \text{ and } \tau_1 <_{\text{lex}} \tau_2 <_{\text{lex}} \cdots <_{\text{lex}} \tau_{\binom{n}{k}}$$

$$g^{\wedge K_k} = \begin{pmatrix} \det g_{\sigma_1 \tau_1} & \det g_{\sigma_1 \tau_2} & \cdots & \det g_{\sigma_m \tau_{\binom{n}{k}}} \\ \vdots & \vdots & \vdots & \vdots \\ \det g_{\sigma_m \tau_1} & \det g_{\sigma_m \tau_2} & \cdots & \det g_{\sigma_m \tau_{\binom{n}{k}}} \end{pmatrix}.$$

Exterior Algebraic Shifting

Definition

The partial (exterior) shift of K by g is

$$\Delta_g^{\mathbb{R}}(K) := \bigcup_k \left\{ \sigma \in \binom{[n]}{k} : \sigma \text{ indexes a column with a pivot of } g^{\wedge K_k} \right\}.$$

Exterior Algebraic Shifting

Definition

The partial (exterior) shift of K by g is

$$\Delta_g^{\mathbb{E}}(K) := \bigcup_k \left\{ \sigma \in \binom{[n]}{k} : \sigma \text{ indexes a column with a pivot of } g^{\wedge K_k} \right\}.$$

(Björner-Kalai '88) For $\mathbb{E}$ sufficiently large we can find $g \in \mathrm{GL}(n, \mathbb{E})$ such that $\Delta_g^{\mathbb{E}}(K) = \Delta^{\mathbb{E}}(K)$. e.g. Take $\mathbb{E} = \mathbb{F}(x_{ij} \mid 1 \leq i, j \leq n)$ and $g = (x_{ij})_{1 \leq i, j \leq n}$

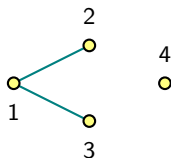
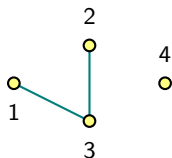
Small Example

For $g = (x_{ij})_{1 \leq i, j \leq 4}$ $K = \{\{1, 2\}, \{2, 3\}, \{4\}\}$, $\mathbb{F} = \text{GF}(2)$.

$$g^{\wedge K_2} = \begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{22}x_{13} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ x_{21}x_{32}+x_{22}x_{31} & x_{21}x_{33}+x_{23}x_{31} & x_{21}x_{34}+x_{24}x_{31} & x_{22}x_{33}+x_{23}x_{32} & x_{22}x_{34}+x_{24}x_{32} & x_{23}x_{34}+x_{24}x_{33} \end{pmatrix}.$$

Which row reduces to,

$$\begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{13}x_{22} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ 0 & x_{11}x_{21}x_{22}x_{33} + x_{11}x_{21}x_{23}x_{32} + x_{12}x_{21}^2x_{33} + x_{13}x_{21}^2x_{32} + x_{12}x_{21}x_{22}x_{31} + x_{13}x_{21}x_{22}x_{31} & x_{11}x_{21}x_{22}x_{34} + x_{11}x_{21}x_{24}x_{32} + x_{12}x_{21}^2x_{34} + x_{14}x_{21}^2x_{32} + x_{12}x_{21}x_{24}x_{31} + x_{14}x_{21}x_{22}x_{31} & x_{11}x_{22}^2x_{33} + x_{12}x_{21}x_{22}x_{32} + x_{13}x_{21}x_{22}x_{32} + x_{12}x_{22}^2x_{31} + x_{13}x_{22}^2x_{31} & x_{11}x_{22}^2x_{34} + x_{12}x_{21}x_{22}x_{32} + x_{14}x_{21}x_{22}x_{32} + x_{12}x_{22}^2x_{4}x_{31} + x_{14}x_{22}^2x_{31} & x_{11}x_{22}x_{23}x_{34} + x_{12}x_{21}x_{23}x_{34} + x_{12}x_{21}x_{24}x_{33} + x_{13}x_{21}x_{24}x_{32} + x_{14}x_{21}x_{23}x_{32} + x_{13}x_{22}x_{24}x_{31} + x_{14}x_{22}x_{23}x_{31} \end{pmatrix}.$$



LU decomposition

Lemma

$\Delta_{gb}(K) = \Delta_g(K)$ for any upper triangular b and any simplicial complex K .

LU decomposition For $g \in GL(n, \mathbb{E})$ with non zero leading principal minors, there exists l lower triangular, u upper triangular such that

$$g = lu .$$

Example

$K = \{\{1, 2\}, \{2, 3\}, \{4\}\}$, and $I^{\wedge K_2}$ reduces to

$$\begin{pmatrix} x_{13}x_{24}+x_{14}x_{23} & x_{12}x_{24}+x_{14} & x_{24} & x_{12}x_{23}+x_{13} & x_{23} & 1 \\ 0 & x_{12}x_{23}x_{24}x_{34}+x_{12}x_{24}^2 & x_{23}x_{24}x_{34} & x_{12}x_{23}^2x_{34}+x_{12}x_{23}x_{24} & x_{23}^2x_{34} & x_{23}x_{34} \\ & +x_{13}x_{24}x_{34}+x_{14}x_{24} & +x_{24}^2 & +x_{13}x_{23}x_{34}+x_{14}x_{23} & +x_{23}x_{24} & +x_{24} \end{pmatrix},$$

Randomized Algorithms

Definition (Randomized Algorithms)

Monte Carlo correct result with probability greater than $1/2$.

Las Vegas if the output can be tested for correctness.

Las Vegas (verification algorithm)

Algorithm Verify u computes the algebraic shift of S

Input: $u \in \mathbb{E}^{n \times n}$, S a k -uniform hypergraph.

Output: true if $\Delta^{\mathbb{F}}(S) = \Delta_u^{\mathbb{F}}(S)$; otherwise false

1: $S' \leftarrow \Delta_u(S)$

2: $\sigma_{\max} \leftarrow \max_{\text{lex}} S'$

3: $T \leftarrow \{\tau \in \binom{[n]}{k} : \tau <_{\text{lex}} \sigma_{\max}\} \mid S' \mid = \mid T \mid + 1 \text{ blah}$

Random Sampling from Small Fields

bad example

taking field extensions helps

bipartite table

blah

Thank you!